MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 14 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, all intermediate algebraic steps, and clearly boxed final answers.
2. In internal choice (**OR**) questions, both options are solved completely with identical step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M, 3M]**.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. If the nth term of an AP is $a_n = 3n - 1$, then its 12th term (a_{12}) is: [1]

(A) 34 (B) 35 (C) 36 (D) 37

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) 35** [0.5 Mark]

Evaluating a_n at $n = 12$:

Formula / Theorem Used: $a_{12} = (3n-1)|_{n=12}$

Justification: [0.5 Mark]

$a_{12} = 3(12) - 1 = 36 - 1 = \mathbf{35}$.

Q2. Which term of the AP $3, 15, 27, 39, \ldots$ will be 132 more than its 54th term? [1]

(A) 64th (B) 65th (C) 66th (D) 67th

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) 65th** [0.5 Mark]

Given: $a = 3$, $d = 15 - 3 = 12$.

Formula / Theorem Used: $a_n - a_{54} = (n - 54)d$

Justification: [0.5 Mark]

$12(n - 54) = 132 \implies n - 54 = \frac{132}{12} = 11 \implies n = 54 + 11 = \mathbf{65}$.

Q3. The sum of the first 12 terms of the AP $8, 11, 14, 17, \ldots$ is: [1]

(A) 294 (B) 300 (C) 312 (D) 324

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) 294** [0.5 Mark]

Given: $a = 8$, $d = 11 - 8 = 3$, $n = 12$.

Formula / Theorem Used: $S_n = \frac{n}{2}[2a + (n-1)d]$

Justification: [0.5 Mark]

$S_{12} = \frac{12}{2}[2(8) + 11(3)] = 6(16 + 33) = 6 \times 49 = \mathbf{294}$.

Q4. Which term of the AP $20, 19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}, \ldots$ is the first negative term? [1]

(A) 27th (B) 28th (C) 29th (D) 30th

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) 28th** [0.5 Mark]

Given: $a = 20$, $d = \frac{77}{4} - 20 = -\frac{3}{4}$.

Formula / Theorem Used: $a_n < 0 \iff a + (n-1)d < 0$

Justification: [0.5 Mark]

$20 - \frac{3}{4}(n-1) < 0 \implies 80 - 3n + 3 < 0 \implies 3n > 83 \implies n > 27\frac{2}{3} \implies n = \mathbf{28}$.

Q5. **Assertion (A):** If $S_n = 5n - n^2$ for an AP, then its common difference is $d = -2$.
Reason (R): When the sum of n terms of an AP is $S_n = An^2 + Bn$, its common difference equals $2A$. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.

(B) Both (A) and (R) are true, and (R) is not correct explanation.

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) Both (A) and (R) are true, and (R) explains (A).** [0.5 Mark]

Comparing coefficients in $S_n = -1 \cdot n^2 + 5n$:

Formula / Theorem Used: $S_n = \frac{d}{2}n^2 + \left(a - \frac{d}{2}\right)n \implies A = \frac{d}{2} \implies d = 2A$

Justification: [0.5 Mark]

Here $a_1 = S_1 = 4$, $S_2 = 6 \implies a_2 = 2 \implies d = -2$, matching $2A = 2(-1) = -2$. Thus both are true and (R) explains (A).

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SECTION B: Very Short Answer Questions — MATHS MANTHAN | (2 × 2 = 4 Marks)

Q6. How many natural numbers between 101 and 999 are divisible by both 2 and 5? **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Identifying the Common Divisor and AP [1 Mark]

Divisible by both 2 and 5 $\iff$ divisible by LCM$(2, 5) = 10$.

Numbers between 101 and 999 divisible by 10: $110, 120, 130, \ldots, 990$ ($a = 110$, $d = 10$, $l = 990$).

Formula / Theorem Used: $a_n = a + (n-1)d$

Step 2: Solving for n [1 Mark]

$$110 + (n-1)(10) = 990 \implies 10(n-1) = 880 \implies n - 1 = 88 \implies \boxed{\mathbf{n = 89}}$$

Q7. Find the AP whose 8th term is 31 and whose 15th term exceeds its 11th term by 16. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $a_8 = 31$ and $a_{15} - a_{11} = 16$.

Formula / Theorem Used: $a_n = a + (n-1)d$

Step 1: Finding d and a [1 Mark]

$(a + 14d) - (a + 10d) = 16 \implies 4d = 16 \implies d = 4$.

Substituting $d = 4$ into $a_8 = a + 7d = 31$: $a + 28 = 31 \implies a = 3$.

Step 2: Forming the AP [1 Mark]

Required AP: **3**, **7**, **11**, **15**, ...

SECTION C: Short Answer Questions — MATHS MANTHAN | (2 × 3 = 6 Marks)

Q8. In an AP, if $a_5 + a_9 = 72$ and $a_7 + a_{12} = 97$, find the AP. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: $a_5 + a_9 = 72$ and $a_7 + a_{12} = 97$.

Formula / Theorem Used: $a_n = a + (n-1)d$

Step 1: Forming Two Linear Equations [1.5 Marks]

$(a + 4d) + (a + 8d) = 72 \implies 2a + 12d = 72 \implies a + 6d = 36$. — (i)

$(a + 6d) + (a + 11d) = 97 \implies 2a + 17d = 97$. — (ii)

Step 2: Solving for d, a, and Writing the AP [1.5 Marks]

Multiplying (i) by 2: $2a + 12d = 72$. Subtracting from (ii): $5d = 97 - 72 = 25 \implies d = 5$.

Substituting $d = 5$ into (i): $a + 6(5) = 36 \implies a = 36 - 30 = 6$.

Required AP: **6**, **11**, **16**, **21**, ...

Q9. Find the sum of all three-digit natural numbers that leave a remainder of 2 when divided by 5. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Identifying the AP and Finding n [1.5 Marks]

Three-digit numbers of the form $5k + 2$: smallest is 102 ($5 \times 20 + 2$) and largest is 997 ($5 \times 199 + 2$).

AP: $102, 107, 112, \ldots, 997$ ($a = 102$, $d = 5$, $l = 997$).

Formula / Theorem Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}(a + l)$

$$102 + (n-1)(5) = 997 \implies 5(n-1) = 895 \implies n - 1 = 179 \implies n = 180$$

Step 2: Computing S_{180} [1.5 Marks]

$$S_{180} = \frac{180}{2}(102 + 997) = 90 \times 1{,}099 = \boxed{\mathbf{98{,}910}}$$

Q10. If the sum of the first m terms of an AP is n and the sum of its first n terms is m $(m \neq n)$, show that the sum of its first $(m+n)$ terms is $-(m+n)$. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Let first term be a and common difference be d. Given $S_m = n$ and $S_n = m$ with $m \neq n$.

Formula / Theorem Used: $S_k = \frac{k}{2}[2a + (k-1)d]$

Step 1: Setting Up Equations for $S_m = n$ and $S_n = m$ [1.5 Marks]

$$\frac{m}{2}[2a + (m-1)d] = n \implies 2am + m(m-1)d = 2n \quad \text{— (i)}$$

$$\frac{n}{2}[2a + (n-1)d] = m \implies 2an + n(n-1)d = 2m \quad \text{— (ii)}$$

Step 2: Subtracting (ii) from (i) and Cancelling $(m-n)$ [2 Marks]

$$2a(m-n) + [(m^2 - n^2) - (m-n)]d = 2(n-m) = -2(m-n)$$

$$(m-n)[2a + (m+n-1)d] = -2(m-n) \implies 2a + (m+n-1)d = -2 \quad (\text{since } m \neq n)$$

Step 3: Evaluating S_{m+n} [1.5 Marks]

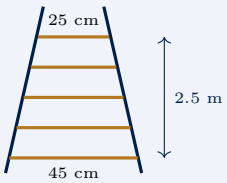
$$S_{m+n} = \frac{m+n}{2}[2a + (m+n-1)d] = \frac{m+n}{2}(-2) = \boxed{-(\mathbf{m+n}) \quad \text{[Hence Proved]}}$$

Q11. A ladder has rungs 25 cm apart. The rungs decrease uniformly in length from 45 cm at the bottom to 25 cm at the top. If the top and the bottom rungs are $2\frac{1}{2}$ m apart, what is the length of the wood required for the rungs? **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Finding the Number of Rungs (n) [2 Marks]

Distance between top and bottom rungs = 2.5 m = 250 cm. Gap between rungs = 25 cm.



Formula / Theorem Used: $n = \frac{\text{Span}}{\text{Gap}} + 1$ and $S_n = \frac{n}{2}(a+l)$

$$n = \frac{250}{25} + 1 = 10 + 1 = 11 \text{ rungs}$$

Step 2: Identifying a and l [1 Mark]

Since lengths decrease uniformly, they form an AP with $a = 45$ cm, $l = a_{11} = 25$ cm, $n = 11$.

Step 3: Total Length of Wood Required [2 Marks]

$$S_{11} = \frac{11}{2}(45 + 25) = \frac{11}{2} \times 70 = 11 \times 35 = \boxed{\mathbf{385} \text{ cm} \quad (\text{or } 3.85 \text{ m})}$$

Q12. CASE STUDY: Sports Day Potato-and-Bucket Relay Race

In a potato race, a bucket is placed at the starting point, which is 5 m from the first potato, and the other potatoes are placed 3 m apart in a straight line. There are ten potatoes in the line. A competitor starts from the bucket, picks up the nearest potato, runs back with it, drops it in the bucket, runs back to pick up the next potato, runs to the bucket to drop it in, and continues until all the potatoes are in the bucket. [5]

(i) Find the total distance run by the competitor to pick up and drop the 1st potato. [1]

(ii) Find the total distance run by the competitor to pick up and drop the 2nd potato. [1]

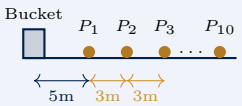
(iii) What is the total cumulative distance the competitor has to run to put all 10 potatoes into the bucket? [3]

— OR —

Calculate the round-trip distance run specifically to retrieve the 10th potato. [3]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Extracted Scenario Data: One-way distances are $5\text{ m}, 8\text{ m}, 11\text{ m}, \ldots$. Round-trip distances form an AP with $a = 10$ m and $d = 6$ m.



Formula / Theorem Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}[2a + (n-1)d]$

Sub-part (i): Round-Trip Distance for 1st Potato [1 Mark]

Running 5 m out and 5 m back: $a_1 = 2 \times 5 = \boxed{\mathbf{10}\text{ m}}$.

Sub-part (ii): Round-Trip Distance for 2nd Potato [1 Mark]

Distance from bucket to $P_2 = 5 + 3 = 8$ m. Round trip: $a_2 = 2 \times 8 = \boxed{\mathbf{16}\text{ m}}$.

Sub-part (iii): Total Cumulative Distance for 10 Potatoes [3 Marks]

Round-trip AP is $10, 16, 22, \ldots$ ($a = 10$, $d = 6$, $n = 10$):

$$S_{10} = \frac{10}{2}[2(10) + (10-1)(6)] = 5(20 + 54) = 5 \times 74 = \boxed{\mathbf{370}\text{ m}}$$

Sub-part (iii) [OR Option]: Round-Trip Distance for 10th Potato [3 Marks]

One-way distance to $P_{10} = 5 + 9(3) = 32$ m; hence round trip:

$$a_{10} = 2 \times 32 = 10 + (10-1)(6) = 10 + 54 = \boxed{\mathbf{64}\text{ m}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★